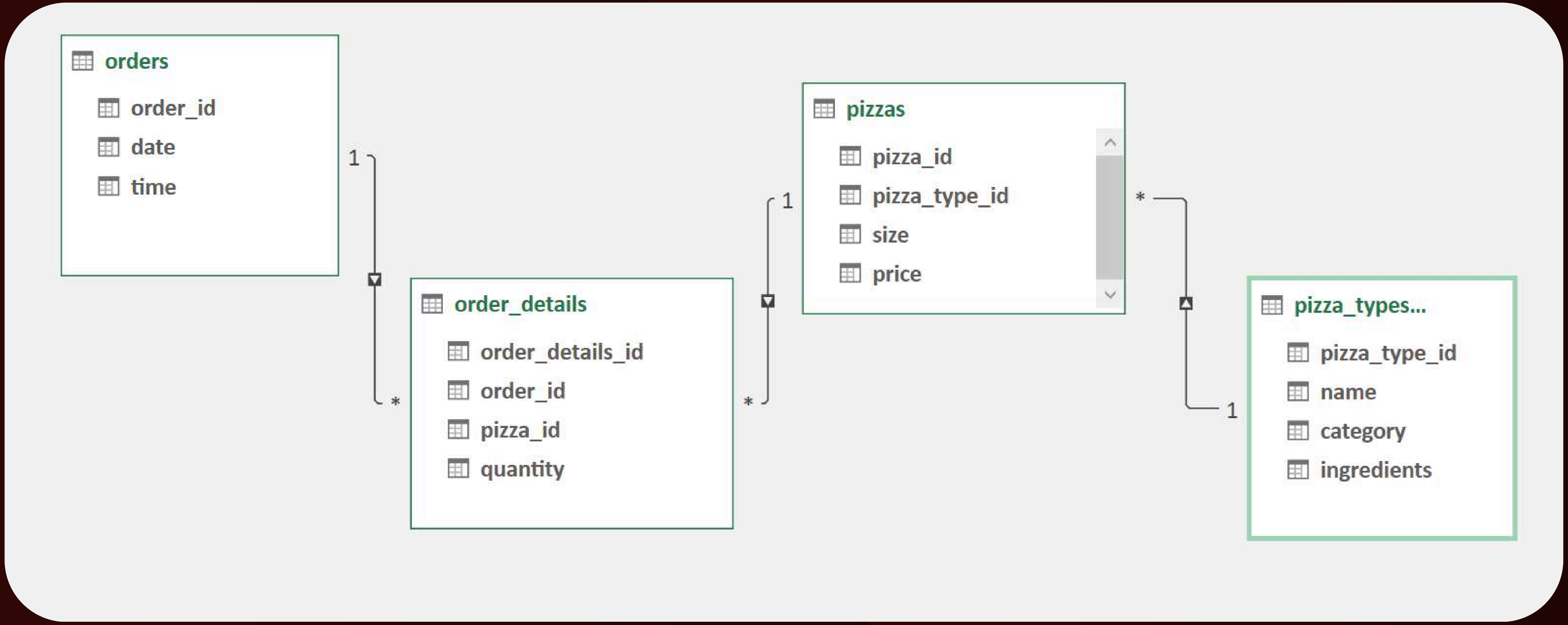
PIZZA HUT SALES ANALYSIS USING SQL

Name: Yallaling Pujari
Tool Used: SQL
Dataset: Pizza Sales Dataset

INTRODUCTION

This project analyzes Pizza Hut sales data using SQL queries. The goal is to understand customer ordering behavior, identify popular pizzas, analyze revenue, and find business insights from the dataset.

PIZZA HUT SALES DATA MODEL



The database schema shows the relationship between Orders, Order Details, Pizzas, and Pizza Types tables. These tables are connected using primary and foreign keys, which helps in performing SQL queries to analyze pizza sales data.

“Now Let’s Explore the Data Using SQL Queries."

In this section, SQL queries are used to answer important business questions, such as:

1. What is the total number of orders placed?
2. What is the total revenue generated from pizza sales?
3. Which pizza size is ordered the most?
4. What are the top-selling pizza types?
5. At what time do customers place the most orders?
6. Which pizza categories generate the highest revenue?
7. These analyses help understand customer behavior, product demand, and business performance.

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED

```sql
SELECT
    COUNT(order_id) AS total_orders
FROM
    orders;
```

Result Grid

total_orders
21350

- This query calculates the total number of orders placed by customers. It helps understand the overall demand for pizzas and the total transaction volume during the analyzed period.

CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES

```sql
SELECT
    ROUND(SUM(orders_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    orders_details
        JOIN
    pizzas ON pizzas.pizza_id = orders_details.pizza_id
```

Result Grid

total_sales
817860.05

- This query calculates the total revenue generated from pizza sales by multiplying the quantity of pizzas ordered with their price. It shows the total income earned by the business from pizza sales.

IDENTIFY THE HIGHEST-PRICED PIZZA.

```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid | Filter Rov

name	price
The Greek Pizza	35.95

- This query identifies the pizza with the highest price on the menu. It helps understand the premium product offered by the restaurant.

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```sql
SELECT
    pizzas.size, COUNT(order_details_id) AS order_count
FROM
    pizzas
        JOIN
    orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid | Filter Ro

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28

- This query finds the pizza size that customers order the most. The result helps understand customer preference and assists in better inventory planning.

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```sql
SELECT
    pizza_types.name,
    SUM(orders_details.quantity) AS total_quantities
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY total_quantities DESC
LIMIT 5;
```

Result Grid | Filter Rows:

name	total_quantities
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

- This query lists the five most popular pizza types based on the total quantity ordered. It helps identify the pizzas that are most preferred by customers.

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```sql
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
```

Result Grid

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

- This query calculates the total quantity of pizzas sold for each category such as Classic, Veggie, Chicken, or Supreme. It helps understand which category contributes the most to overall sales.

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```sql
SELECT
    HOUR(order_time), COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time)
ORDER BY order_count DESC;
```

Result Grid | Filter Rows:

HOUR(order_time)	order_count
18	2399
17	2336
19	2009
16	1920
20	1642
14	1472
15	1468
11	1231

- This query analyzes the number of orders placed during each hour of the day. It helps identify peak business hours when customer demand is highest.

JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.


```
select category ,count(name) from pizza_types
group by category;
```

Result Grid | Filter Rows

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

- This query shows how pizza orders are distributed across different pizza categories. It helps understand which category is most popular among customers.

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```sql
SELECT
    ROUND(AVG(quantity), 0) as avg_pizzas_orders_par_day
FROM
    (SELECT
        orders.order_date, SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY order_date) AS order_quantity;
```

Result Grid | Filter Rows

avg_pizzas_orders_par_day
138

- This query calculates the average number of pizzas ordered per day. It helps measure the daily sales performance of the restaurant.

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```sql
SELECT
    pizza_types.name,
    SUM(orders_details.quantity * pizzas.price) AS revenu
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenu DESC
LIMIT 3;
```

Result Grid | Filter Rows:

name	revenu
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

- This query identifies the three pizza types that generate the highest revenue. It helps determine the most profitable pizzas in the menu.

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```sql
SELECT
    pizza_types.category,
    ROUND((SUM(orders_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(orders_details.quantity * pizzas.price),
                                2) AS total_sales
                FROM
                    orders_details
                        JOIN
                    pizzas ON pizzas.pizza_id = orders_details.pizza_id)) * 100,
            2) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid | Filter

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

- This query calculates the percentage contribution of each pizza type to the total revenue. It helps understand which pizzas contribute the most to the business income.

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
select order_date,
sum(revenu) over(order by order_date) as cum_revenu
from
(select orders.order_date,
sum(orders_details.quantity*pizzas.price) as revenu
from orders_details join pizzas
on orders_details.pizza_id= pizzas.pizza_id
join orders
on orders.order_id=orders_details.order_id
group by orders.order_date)as sales;
```

order_date	cum_revenu
2015-01-01	2713.850000000
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7

- This query analyzes how revenue accumulates over time. It helps track sales growth and understand revenue trends.

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```sql
select name,revenue
 from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category,pizza_types.name,
sum((orders_details.quantity)*pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on pizzas.pizza_id=orders_details.pizza_id
group by pizza_types.category,pizza_types.name ) as a) as b
where rn <=3;
```

esult Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

- This query identifies the top three revenue-generating pizzas within each category. It helps understand the best-performing pizzas in every category.

CONCLUSION

- Large size pizzas were ordered the most.
- Certain pizza types generated higher revenue.
- Peak ordering hours helped identify busy periods.
- SQL analysis helps businesses improve sales strategy